MOBILE DJ / PRODUCT BLUEPRINT

DOCUMENT 10 OF 19

Delivery roadmap and agent playbook

A new touch-first performance instrument.
Professional audio, liquid glass and mechanical craft.



DESIGN HANDOFF BASELINE - PLANNING ONLY

29 October 2026 | Crystal Current selected; S26 Ultra test target

Independent of the previous Mixline preview. Targets are not measured results.

Version 1.0 | 29 October 2026 | Design handoff baseline; implementation and deployment not started

1. Delivery strategy

Build the product through evidence gates. The objective is the full agreed professional mobile DJ product; a gate is a proof of readiness for the next investment, not permission to silently remove advanced requirements. This planning task ends with a reconciled design handoff; Document 19 provides the next agent's entry point. No further owner answer is needed to start design. No application, repository deployment, provider account, paid licence, or public release has been created.

The audience, C Crystal Current, Galaxy S26 Ultra first device, ParDay armed-assisted capability and personal-tool delivery are selected. The owner is open to a later GitHub source release for community home labs; publication and licence remain decisions for that stage. The latest decision is local first: develop the phone-native app and on-device acquisition path without a required server. The earlier on-device versus self-hosted comparison is retained for a later optional AWS expansion, with no measured winner. The future AWS planning target is approximately US$35 per month, excluding ParDay AI-provider charges; this is not a hard billing ceiling or deployment/spending authorization. Account, region, service and resources wait for a concrete expansion proposal. The critical path is exact route/build capture, audio feasibility, one real end-to-end mix, interaction validation, breadth on the same engine, then release certification. Visual components and asset exploration can run alongside audio work after they share the same interaction contracts. Source-access feasibility can run early, but an unavailable integration or home-lab service must not become a hidden dependency in the local engine.

2. Canonical delivery gates

Gate	Deliverable	Exit evidence	Dependency
G0 - Decision baseline	Recorded owner choices, local-first sequence, selected visual direction and versioned requirements	Design can start from Document 19; proposed details and later evidence gates are labelled	Current planning dossier
G1 - Audio feasibility	Two native decks, candidate stretch engines, sample-clock scheduling, real output routes	Hardware timing captures, warm-soak logs, DSP listening review, licence route, fail/revise decision	G0; capture device/route and any paid experiment envelope when needed
G2 - Working vertical slice	Import two real tracks, prepare, load, cue, transition, save and restore	Demonstration on the named real target device; UI and audio are connected; no simulated waveform or meter values	G1 engine and bridge contracts
G3 - Core performance beta	Reliable two-deck mixer, library, grids, cues, loops, FX, sampler, recording eligibility and recovery	Traceable functional tests, accessibility review, owner workflow sessions, sustained performance profile; optional independent DJ review	G2 and assets for core components
G4 - Advanced performance	Four-deck workspace, prepared stems, richer sampling/remix features, gated live stems	Per-profile sustained tests, stems quality/alignment, advanced interaction review	G3 and explicit device tiers
G5 - Source/provider and ParDay integrations	Source adapters, requested on-device link acquisition and selected assistant capability	Technical proof plus separate provider-access, content-rights, distribution, token/expiry and privacy evidence; local authority and takeover tests	Adapter-specific evidence and personal scope; cloud extension cases deferred; official API not a universal technical prerequisite
G6 - Release candidate	Complete selected personal-release matrix, final assets, installation package and operating documentation	No release-blocking failures, signed build reproducibility, licence inventory, support readiness	Required portions of G3-G5 and resolved launch scope
G7 - Personal installed release	Verified signed local build installed on the owner's S26 Ultra	Install/upgrade, backup/restore and offline/outage evidence; working local setup instructions	G6 and owner-directed installation; AWS/home-lab setup and public-store approval are not prerequisites
G8 - Maintenance and community readiness	Personal updates, regression checks and optional source/release preparation	Reproducible setup, licence/asset manifest, secret-free examples and a tested recovery path	G7; any GitHub publication remains a separate action

G4 and G5 can overlap. AWS and service-specific home-lab tests are deferred; local cases remain required. Complete personal release includes all selected advanced, source, ParDay and specialist modules. Lighting stays excluded. Run specialist feasibility in G1, integration in G4/G5 and release evidence in G6. Report blocked formats/providers explicitly; G3 is not the finished product.

3. Work ownership

Agent/role	Owns	Must not decide alone	Required handoff
Product and integration lead	Requirements, priorities, decision ledger, release assembly	Scope removals, claims of user approval	Updated traceability and unresolved decisions
Native audio/DSP engineer	Engine clock, buses, transport, DSP graph, routing adapters	Music rights, public performance promises	Versioned engine API, timing data, audio fixtures
Android platform engineer	Lifecycle, foreground playback, files, audio focus, device routes	Ignoring system interruption rules	Device/event matrix and instrumentation
UI/interaction engineer	Compose screens, touch capture, accessibility semantics, state presentation	DSP truth, unsupported device capability	Rendered screens and input-state tests
Visual/technical artist	Tokens, vector controls, materials, iconography, motion	Adding expensive effects to live path	Asset manifest, source masters, fallback variants
Library/data engineer	Track identity, crates, search, analysis jobs, migrations, backups	Overwriting user cues during reanalysis	Schema migrations and conflict/recovery examples
Analysis/ML engineer	Beat/key analysis, waveform pyramids, stem generation	Equating a model licence with all weight/data rights	Reproducible model manifest, quality/thermal results
Music integration engineer	Capability resolver, contracted provider adapters	Treating public APIs as DJ authorization	Per-provider rights/capability table and tests
Cloud/home-lab service engineer - later extension	Deferred comparative acquisition worker, pairing, optional AI gateway, retention and updates	Making a server a local-release prerequisite, selecting a winner without evidence or making live audio server-dependent	Reviewable expansion proposal, then reproducible deployment and measured cost/outage/deletion behavior
ParDay AI engineer	Provider adapters, typed proposals, source-filtered context, authority validator	Giving model output direct audio authority or unreviewed data egress	AI-001-024 fixtures, cost/retention settings, rejected-action evidence
Audio/video and plugin engineer	Video clock/composition, karaoke, broadcast encoders, real plugin host	Claiming unsupported mobile formats or bypassing distribution rules	Document 15 compatibility profile and AV evidence

Agent/role	Owns	Must not decide alone	Required handoff
QA/performance engineer	Fixtures, traces, device profiles, soak and recovery tests	Accepting visual simulation as functional proof	Evidence bundle mapped to requirement IDs
Security/release engineer	Dependency review, signing, privacy, distribution checks	Sending messages, buying licences, publishing without authorization	Release checklist and verifiable build provenance
DJ reviewer and accessibility reviewer	Actual workflow and listening acceptance	Certifying untested devices or audiences	Observations, failures, and recommended changes

One person can fill several roles, but independent review is needed for the callback, route changes, capability enforcement, and release claims. AI agents accelerate bounded implementation; their output is not evidence that a phone sounds or behaves correctly.

4. Proposed repository organization

Create these locations when implementation is authorized. They are a starting convention, not existing inspected files:

```
app-android/                Native app, lifecycle, navigation
core/domain/                Shared IDs, commands, immutable state
core/data/                  Room, migrations, import and search
audio/engine/               Platform-independent C++ mix graph
audio/android/              Oboe and platform routing adapter
audio/dsp-adapters/          Replaceable licensed DSP interfaces
analysis/                   Beat/key/waveform/stem workers
features/performance/        Deck, mixer, waveform, pads and FX
features/library/            Import, crates, preparation and search
features/routing/            Output and headphone setup
integrations/                Source adapters and capability policies
services/acquisition/        Deferred optional owner-operated worker
deploy/                      Deferred AWS/home-lab examples, no secrets
benchmarks/acquisition/      Common comparison protocol and results
docs/homelab/                Optional future service setup and recovery
assets/source/              Editable masters and licence evidence
assets/runtime/             Optimized production exports
contracts/                  Versioned public interfaces and schemas
tests/fixtures/             Rights-cleared, deterministic inputs
tests/device/               Hardware configurations and evidence
docs/                       Approved requirements and decisions
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Keep UI feature modules dependent on domain contracts rather than each other's implementation. Only the audio owner changes the callback graph. UI and integration agents can propose API changes through the integration lead.

5. First implementation assignments

Task A01 - engine and route spike. Read Documents 05, 06, and 09. Build a bounded native audio harness with two local PCM fixtures, one sample clock, gain controls, equal-power and linear crossfades, stream restart, and independent master/cue tests where hardware permits. Output timing traces and a reproducible hardware configuration. Stop if the proposed route cannot carry the required channels; do not simulate cue separation.

Task A02 - performance interaction prototype. Start with Document 19, then read Documents 03, 04, and 08. Build representative portrait and landscape screens containing actual touch targets, deck focus,

browse sheet, and control states. Early fixtures must be visibly identified as prototype data. Validate protected seek versus scratch, playing-deck replacement, hidden-deck observability, and rotation. Integrate the real A01 engine before claiming G2.

Task A03 - durable local library. Read Documents 02 and 07. Import a permitted local folder/file set; persist identity and user annotations; handle revoked URI access and missing media; analyze asynchronously. Demonstrate that reanalysis preserves edited cues/grids and that a failed import cannot damage the previous crate.

Task A04 - DSP candidate comparison. Compare time-stretch/key-shift engines using the same files, tempo range, output format, and hardware. Include transients, vocals, sustained harmony, silence, variable tempo, and rapid cue/seek changes. Report licensing, integration time, reported latency, memory, CPU, and listening results. Do not compare marketing claims as benchmark results.

Task A05 - assets and design tokens. Produce the representative knob, fader, pad, meter, waveform markers, and glass surface at multiple states and scales. Deliver source plus optimized export and an opaque fallback. Validate those components before making every screen.

Task A06 - recovery and recording. Exercise activity recreation, route removal, focus loss, full storage, recording permission changes, and process restart. Record what is preserved and what requires user action. A force-stopped or killed process cannot be promised uninterrupted audio.

Task A07 - ParDay deterministic harness. Read Documents 07, 13 and 14. Use synthetic eligible fixtures before live API access. Prove context provenance, typed proposals, user scope, state freshness, idempotency and manual takeover. Only then attach a replaceable provider adapter. Cloud-model latency must never enter the audio deadline.

Task A08 - on-device link acquisition first. Read Documents 07, 14 and 16 plus contracts/acquisition_comparison_protocol.json. Implement the native intake/queue and validated publication contract with controlled fixtures. Prove an actual on-device candidate for the requested source path, record source/network conditions and retain the baseline data for later comparison. Preserve requested and actual-source identities; include unofficial routes as concrete candidates rather than replacing them with discovery-only handoff. Report technical, access/content and distribution evidence separately. A mocked download is not capability evidence. If an intended local adapter is infeasible or inadmissible, report the specific blocker; do not silently require AWS or mark FR-049 complete. Documents 17 and 18 guide the deferred service extension, not this task's prerequisites.

Task A09 - advanced module feasibility. Read Document 15. Verify video/audio sync under seek/scratch, karaoke microphone/output, an independent encoder failure and one real third-party Android plugin. Pin supported format/distribution choices and workload budgets. Report a plugin-hosting blocker if only built-in effects work.

Task A10 - local personal installation and recovery. Read Documents 11 and 17. Specify stable private signing, local install/upgrade, configuration/secret separation and backup/restore. Rehearse a prepared set with network and optional AI unavailable. Verify the phone-native release without requiring an AWS or home-lab service. Preserve versioned worker contracts for later expansion. A future community release needs buildable code, dependency/asset notices and truthful supported profiles; no music library, credentials or signing keys belong in it.

Task A11 - optional AWS expansion, deferred. Activate after a usable local baseline exists and the owner commissions the expansion. Read Documents 16-18 and the deployment/cost contracts. First identify the measured bottleneck or useful offload outcome, then prepare a concrete source-path trial, region/service proposal, current cost estimate within the approximately US$35/month target, pairing, retention and teardown. Keep AI charges separate. Compare D and AWS with the common protocol before selecting supported cloud behavior. No AWS account, deployment or spending is required for A01-A10. Optional LAN/remote home-lab profiles remain portability references.

These assignments define testable work; they do not authorize starting it during this planning session.

6. Required task envelope

Every coding assignment must state its objective, requirement IDs, owner, allowed paths, dependency versions, input/output schema, non-goals, failure behavior, acceptance cases, device requirements, and definition of done. Work ends with a handoff containing changed files, interface changes, tests actually run, supporting artifacts, untested cases, known limitations, and the next dependency.

Use this compact prompt pattern:

```
Implement task [ID] against dossier version [VERSION].
Read [DOCUMENTS] and the current decision ledger first.
Own only [PATHS]; coordinate any contract change with [OWNER].
Satisfy [REQUIREMENT IDS] and execute [ACCEPTANCE IDS].
Use real engine state for any live signal or transport claim.
Report evidence, device/build identity, unresolved risks, and handoff.
Do not change product scope or present provisional choices as approved.
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Agents must distinguish proposed paths from inspected code and report missing dependencies candidly. They must not work around access controls, embed credentials, or invent successful tests. Provider outreach, paid procurement, app-store publication, and external messages need the applicable user authorization.

7. Integration and change control

Create one owner per shared contract. Give each parallel agent a bounded branch or worktree. Merge interface changes before dependent feature branches. A change to source timing, stem latency, persistent IDs, routing, or provider capability semantics requires an updated contract version and targeted migration/regression review.

Priority order for conflicting instructions: current user decisions, explicit exclusions, approved decision ledger, approved requirements, approved detailed specifications, then implementation suggestions. This baseline records user-confirmed product choices separately from proposed implementation and acceptance details. It is not authorization to build or deploy. A newer file is not automatically authoritative if its decisions were never accepted.

Require a review when a change can alter audible output, replace user data, change capability enforcement, or weaken a performance gate. For ordinary styling corrections, use the existing visual acceptance criteria without inventing another approval ceremony.

Maintain a requirement matrix with columns ID, statement, gate, owner, implementation reference, acceptance case, evidence, state, and blockers. Valid states are proposed, approved, building, verified, blocked, or explicitly deferred. A checklist tick without evidence cannot move an item to verified.

8. Schedule and cost estimation

No reliable fixed delivery date can be given before G1 and the team/SDK decisions. A useful estimation model is: effort per work package + integration/review effort + device validation + contingency, divided by effective parallel capacity and constrained by the dependency graph. Do not divide the total by the number of AI agents and call the result a schedule.

An illustrative planning allocation for a future budget is 10% discovery and experiments, 30% engine/platform integration, 20% UI and assets, 15% data/analysis, 15% QA/device certification, and 10% release/support preparation. These percentages are an assumption for estimating, not measured costs. Streaming agreements, advanced ML, iOS launch parity, or video can substantially change the allocation.

Obtain separate cost lines for DSP licences, stem-model/weight rights, music partnerships, developer accounts, test devices, audio interfaces, commissioned assets, rights-cleared test music, optional hosting, and human review. Record the currency, vendor quote date, taxes, renewal basis, and cancellation/export

terms. Do not silently spend money to satisfy a specification.

9. Demonstration and definition of done

A milestone demo should show a real track import, preparation change, deck load, cue audition where supported, a transition, a performance gesture, background/rotation behavior, and session recovery. Show the device, audio route, build identifier, and any restricted feature state. Record audio independently where possible so visible animations cannot conceal a dropout.

G6 is complete only when the agreed launch requirements pass functional, sonic, visual, accessibility, privacy, rights, and endurance gates; assets have provenance; personal installation/update/backup instructions describe real limitations; and diagnostics can explain a route or source problem. Every material failure either blocks release or has an explicit, user-accepted scope decision. No claim of "all professional features" is made until the corresponding launch matrix is verified.

10. Design readiness and later decisions

Start design from Document 19, using A02/A05 and the local import-to-mix workflow. No blocking owner question remains for that step. Use the recorded C direction, phone-native Android target, S26 Ultra first device, full professional scope, ParDay authority rules and local-first sequence. The two-deck integration view is a first slice; it does not remove four-deck capability or other selected modules.

Capture actual build/audio-route evidence during G1, select DSP and source mechanisms through measured feasibility work, and choose the first model/provider and AI spending limit before a paid integration. Naming, typed-first voice defaults, wider device support and optional publication/licence choices can be resolved when they affect concrete work. Team capacity and estimates remain unknown, so no fixed delivery date is promised.

AWS is later, with an approximately US$35/month planning target excluding AI-provider charges. Its account, region, service, current quote and operating choices belong to A11, not a questionnaire before local design. Document 12 separates confirmed decisions from these later gates; Documents 16-18 preserve the requested acquisition, portability and future cloud work without making them prerequisites to the local interface.